Cracking the Code: Data Structures and Algorithms (DSA) in JavaScript - II

Marks - 150

Note: Attempt this assignment after studying pre-recorded content and after attending Live lectures of this course.
Create your Leetcode Profiles and solve these questions there. Share your solved questions link along with the Time complexity and Space Complexity of your solution in a doc when you submit your assignment.
Submit the optimised solution for all the questions.

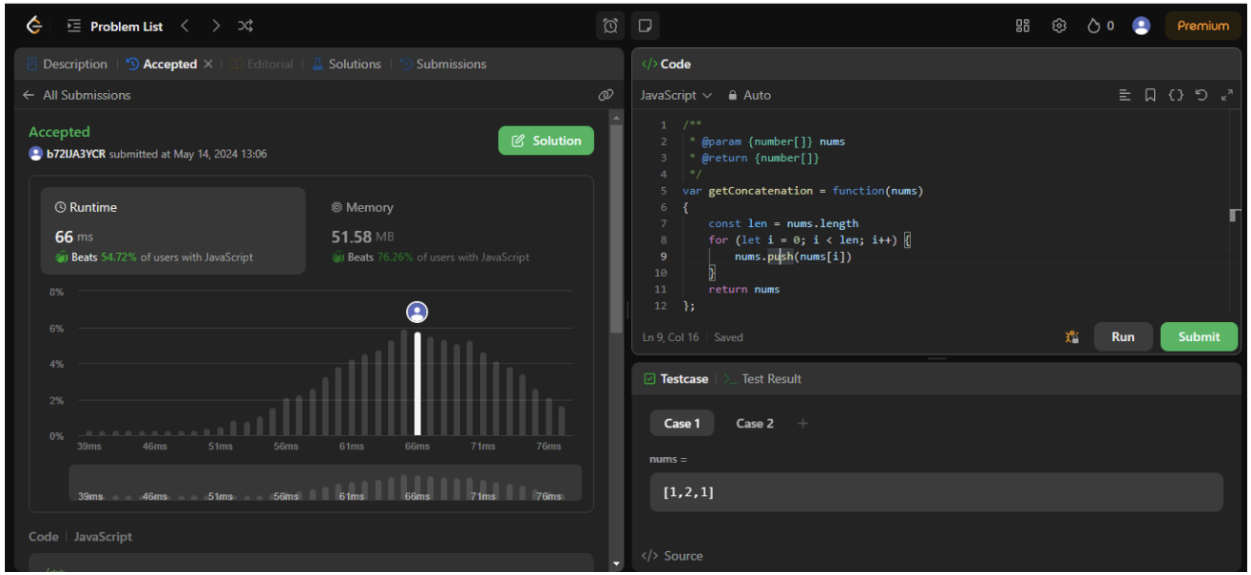
WARNING !! Don't try to copy from somewhere else. We can call any student randomly to explain their solutions and if we find the discrepancy, you will be provided 0.

For Example :
Question was https://leetcode.com/problems/concatenation-of-array/description/

Solution Link -
https://leetcode.com/problems/concatenation-of-array/submissions/1257621340/

Time Complexity: O(n)
Space Complexity: O(1)

Add screenshot:



Description:
Time Complexity: O(n)
Iterating to the entire n-sized array and for each array element performing O(1) operation of push.
Space complexity: O(1)
No Extra space required for the Program Execution

1. Given the head of a singly linked list, return true if it is a palindrome or false otherwise Leetcode 234 (20 marks)

2. Given the head of a linked list, remove the nth node from the end of the list and return its head. Leetcode 19 (20 marks)

3. Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid. Valid Parantheses (20 marks)

4. The next greater element of some element x in an array is the first greater element that is to the right of x in the same array. Next Greater Element (20 marks)

5. Given a binary tree, find its minimum depth. The minimum depth is the number of nodes along the shortest path from the root node down to the nearest leaf node.
Minimum Depth of Binary Tree (25 marks)

6. Given the root of a binary tree, return the postorder traversal of its nodes' values. Binary Tree Postorder Traversal (20 marks)

7. There is a singly-linked list head and you have to delete a node in it.
Leetcode 237 (25 marks)